

FYS-STK3155/4155 September 28



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$$\hat{E} = \arg \min_{E \in \mathbb{R}^p} C(E)$$

$$C(E) = - \sum_{i=0}^{n-1} \left[y_i' (E_0 + E_1 x_i) - \ln(1 + e^{E_0 + E_1 x_i}) \right]$$

$$z_i = E_0 + E_1 x_i$$

$$\left(z_i = E_0 + \sum_{j=1}^p E_j x_{ij} \right)$$

$$= \sum_{i=0}^{n-1} C_i(\theta)$$

$$\frac{\partial C_i(\theta)}{\partial \theta_j} = \frac{\partial C_i'(\theta)}{\partial z_i'} \frac{\partial z_i'}{\partial \theta_j}$$

$$C(\theta) = - \sum_{i=0}^{n-1} \left[y_i' (\theta_0 + \theta_1 x_i') \right. \\ \left. - \ln (1 + e^{\theta_0 + \theta_1 x_i'}) \right]$$

$$z_i' = \theta_0 + \theta_1 x_i'$$

$$C(\theta) = - \sum_{i=0}^{n-1} \left[y_i' z_i' - \ln (1 + e^{z_i'}) \right]$$

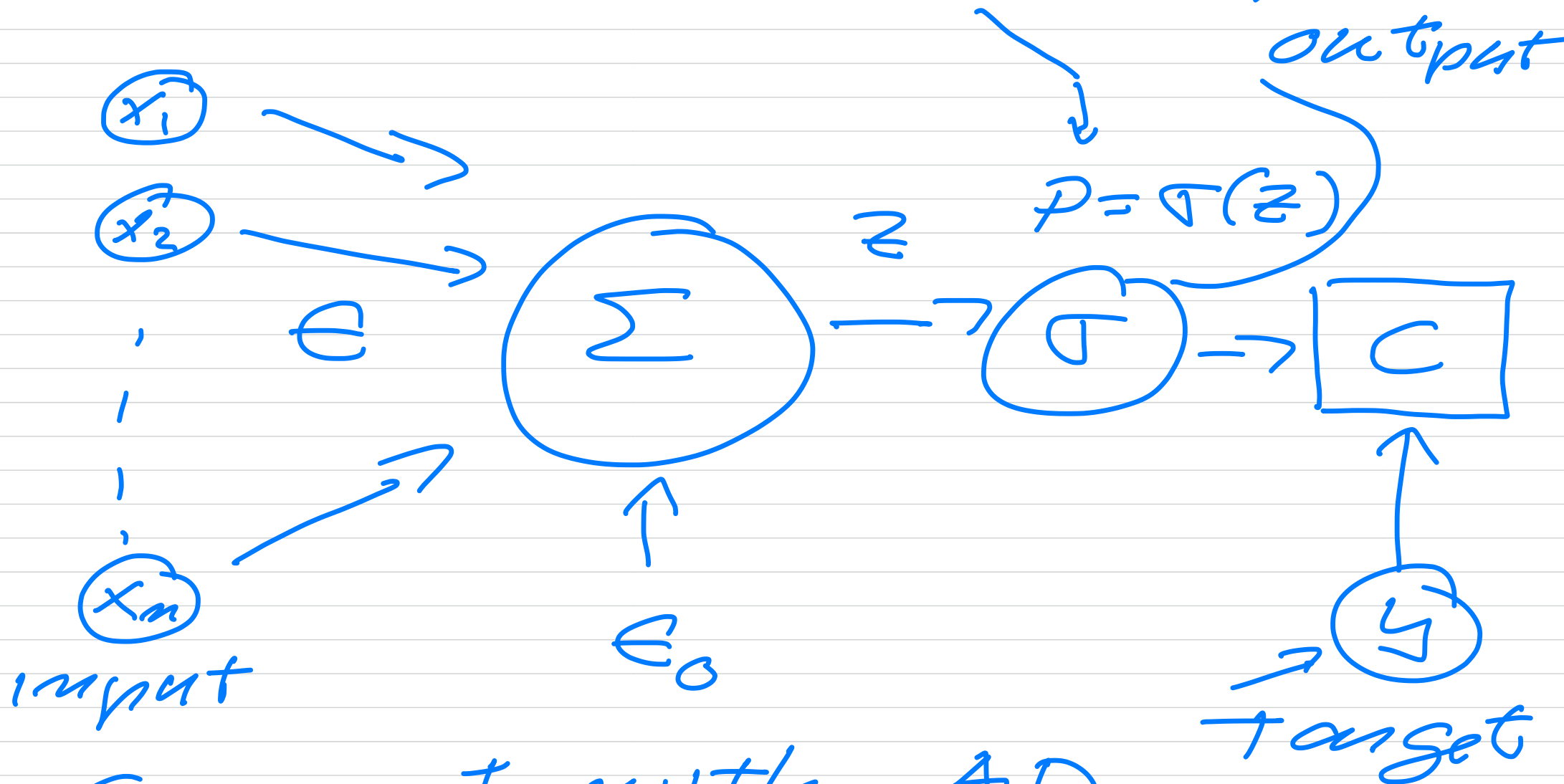
$$\frac{\partial C_i}{\partial \theta_j} = \left(-y_i' + \frac{e^{z_i'}}{1+e^{z_i'}} \right) \underbrace{\frac{\partial z_i'}{\partial \theta_j}}_{x_{ij}'}$$

$$z_i' = \theta_0 + \sum_{j=1}^{p-1} \theta_j x_{ij}'$$

$$z_i' = \theta_0 + \theta_1 x_i'$$

$$\left(\frac{p_i' - y_i'}{e^{z_i'} / (1+e^{z_i'})} \right) x_{ij}' \Rightarrow$$

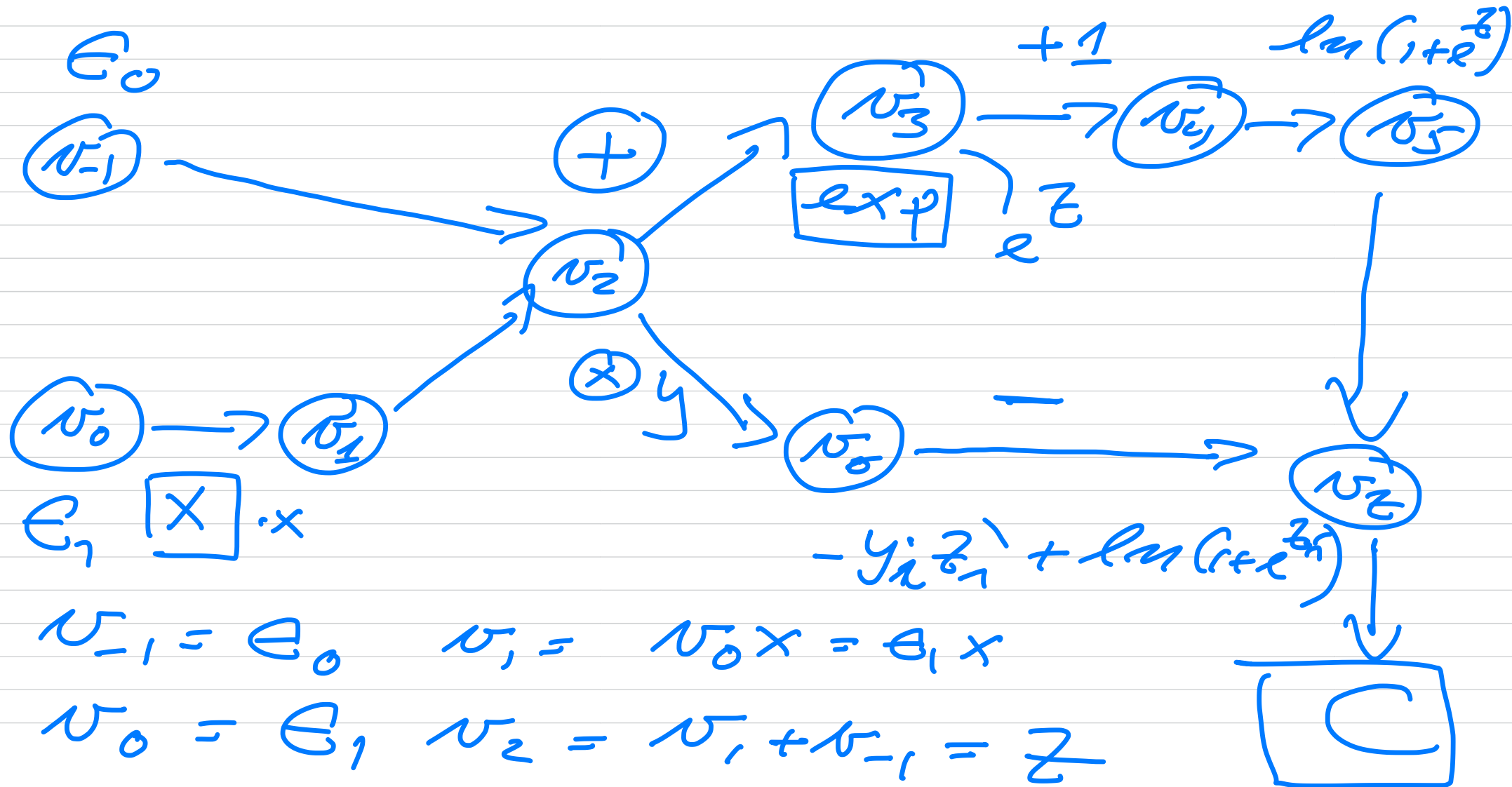
$$\vec{\nabla}_{\theta} C(\theta) = X^T (\vec{p} - \vec{y})$$



Connect with AD
simplify to just b_0 and b_1

$$(i) E_0 + \Theta_1 x = z$$

$$(ii) -\sum (y_i z_i - \ln(1 + e^{z_i}))$$



$$\partial v_1 / \partial v_0 = x$$

$$\partial v_2 / \partial v_5 = \underline{1}$$

$$\partial v_2 / \partial v_1 = \underline{1}$$

$$\partial v_2 / \partial v_0 = \underline{-1}$$

$$\partial v_3 / \partial v_2 = v_3$$

$$\frac{\partial v_4}{\partial v_3} = \underline{1}$$

$$\partial v_5 / \partial v_4 = 1/v_4$$

$$\partial v_6 / \partial v_2 = y$$

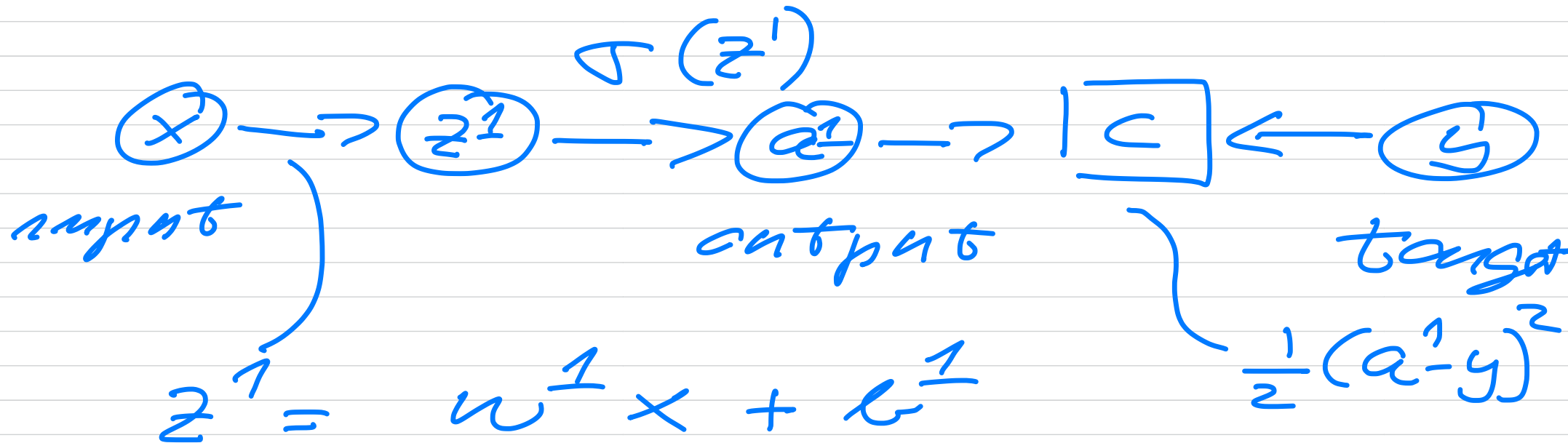
$$v_3 = e^{v_2} = e^z \quad \wedge \quad v_6 = y, v_2 = y, z$$

$$v_4 = 1 + v_3 \quad \wedge \quad v_2 = \zeta_1 = v_5 - v_6$$

$$v_5 = \ln v_4 = \ln(1 + e^z)$$

Warming up towards a full NN code

(1) scalar in / scalar out



parameters $\{w^1, b^1\}$

$$\frac{\partial C}{\partial w^1}$$

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$$\frac{\partial C}{\partial b^1}$$

$$\frac{\partial C}{\partial w^1} = \frac{\partial C}{\partial a^1} \frac{\partial a^1}{\partial z^1} \frac{\partial z^1}{\partial w^1}$$

$$\left[\begin{array}{l} z^1 = w^1 x + b^1 \\ C = \frac{1}{2} (a' - y)^2 \\ a^1 = \sigma(z^1) \end{array} \right]$$

$$= (a' - y) \frac{\partial a^1}{\partial z^1} x = \delta^1 x$$

$$\delta^1 = (a' - y) \sigma'(z^1)$$

$$\frac{\partial C}{\partial b^1} = \underbrace{\frac{\partial C}{\partial a^1} \frac{\partial a^1}{\partial z^1}}_{\delta^1} \frac{\partial z^1}{\partial b^1}$$

$$z^1 = w^1 x + b^1$$

$$\frac{\partial z^1}{\partial b^1} = 1 \Rightarrow$$

$$\frac{\partial C}{\partial w^1} = \delta^1 x \quad \wedge \quad \frac{\partial C}{\partial b^1} = \delta^1$$

gradient descent

$$w^1 \leftarrow w^1 - \eta \delta^1 x$$

$$b^1 \leftarrow b^1 - \eta \delta^1$$

Algorithm:

- choose $C(w, b) = \frac{1}{2} (a^1 - y)^2$

- set η

- initialize w^1, b^1

- have x and y

For while $\frac{1}{2} (a^1 - y)^2 \geq \epsilon$

- Forward pass

$(x) \rightarrow (z^1) \rightarrow (a^1) \rightarrow a^1$
compare with y

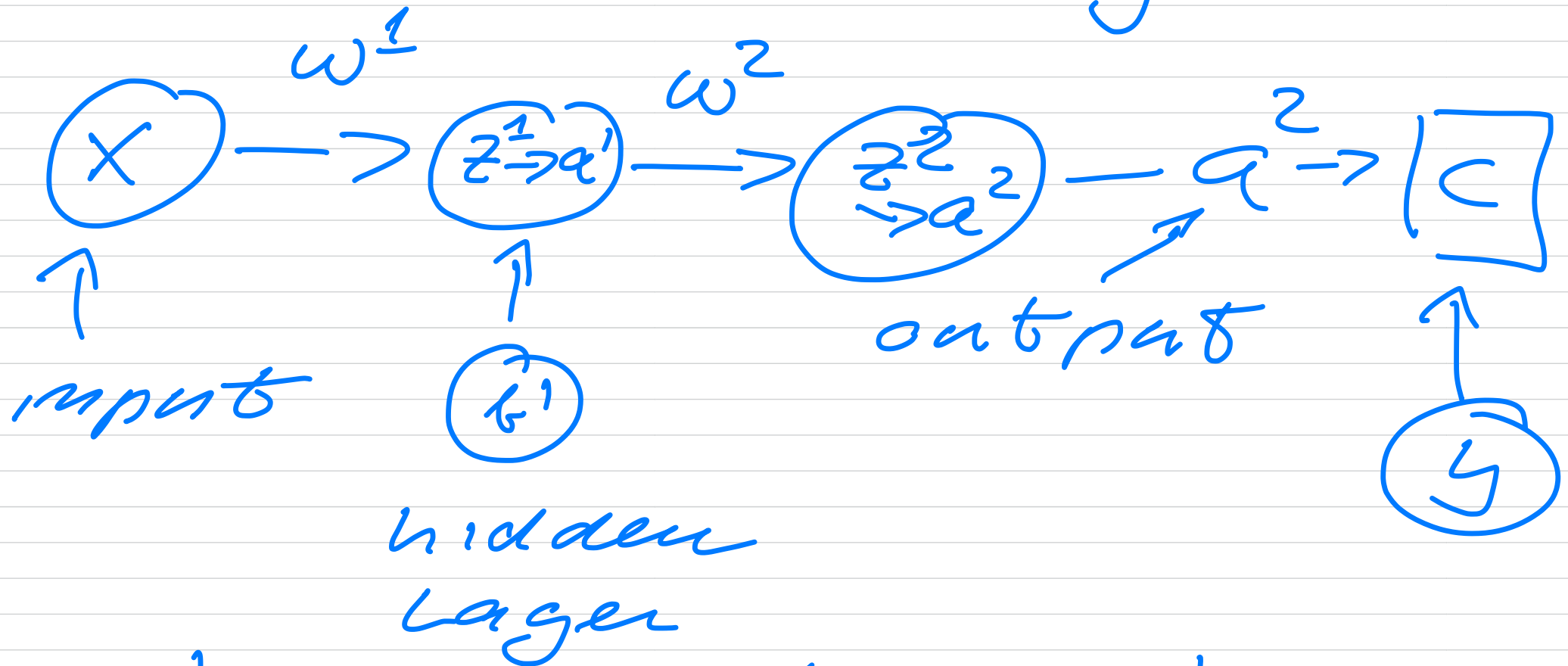
- Backpropagate

$$w^1 \leftarrow w^1 - \delta^1 x$$

$$b^1 \leftarrow b^1 - \delta^1$$

Continue till convergence
Criterion is reached

one hidden layer



$$z^1 = w^1 x + b^1 \quad \wedge \quad a^1 = \sigma(z^1)$$

$$z^2 = w^2 a^1 + b^2 \quad \wedge \quad a^2 = \sigma(z^2)$$

$$C = \frac{1}{2} (a^2 - y)^2$$

$$\frac{\partial C}{\partial u^2} = \frac{\partial C}{\partial a^2} \frac{\partial a^2}{\partial z^2} \frac{\partial z^2}{\partial u^2}$$

$$= \underbrace{(a^2 - 5) \varphi'(z^2)}_{\delta^2} \cdot e^1$$

$$\frac{\partial C}{\partial h^2} = \delta^2 = \frac{\partial C}{\partial a^2} \frac{\partial a^2}{\partial z^2} \frac{\partial z^2}{\partial h^2}$$

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$$\frac{\partial C}{\partial w^1} = \frac{\partial C}{\partial a^2} \frac{\partial a^2}{\partial z^2} \frac{\partial z^2}{\partial w^1} \quad \times$$

$$\frac{\partial z^2}{\partial w^1} = \frac{\partial z^2}{\partial a^1} \frac{\partial a^1}{\partial z^1} \frac{\partial z^1}{\partial w^1} \quad \left. \begin{array}{l} w^2 \\ \delta^1(z') \end{array} \right\}$$

$$z^2 = w^2 a^1 + b^2 = w^2 \sigma(z') + b^2$$

$$\delta^1 = \delta^2 w^2 \sigma'(z')$$

$$z' = w^1 x + b^1$$

$$w^2 \leftarrow w^2 - \gamma \sigma^2 a^1$$

$$b^2 \leftarrow b^2 - \gamma \sigma^2$$

$$w' \leftarrow w' - \gamma \sigma^1 x$$

$$b' \leftarrow b' - \gamma \sigma^1$$